

Lesson 19 — Draw the Sequence First

Matches LEGO Teacher Portal lesson: "Submarine Adventure".

Objective (I can...): I can draw my sequence in pictures first (picture pseudocode), then build it. **CS concept:** Sequences with simple picture pseudocode (plan in pictures, then code) · **Builds on:** Lesson 18 (sequences that solve problems).

AI Connection (2 min, read aloud): Engineers sketch an AI's steps as little pictures before they code — a picture plan. *Ask the group:* "Why might drawing your plan help before you build it?"

Materials: LEGO CS & AI kit + device per pair, a picture "storyboard" strip (4-5 empty boxes) per pair.

Model reuse: Sequences model from Lesson 16 — code only.

60-minute flow - Warm-up (10): "Storyboard it" — pairs draw a 4-box picture plan for a tiny routine (make a sandwich) and point through it. Pictures can be a plan. - **Concept + demo (8):** Coach draws a 4-picture plan in the boxes, then matches each box to a picture block while kids watch — picture plan to code. - **Build/Code (25):** Pairs draw a 4-5 box picture plan for a sequence, then code it box-by-box on the built model and run to check code matches the plan. **Make-it-yours beat:** decorate the model to fit the plan. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Change one picture in the plan and update the matching block so plan and code still agree. - **Share + tidy-up (10):** Pairs hold up their picture plan next to the running program and point box-to-block. Model stays built.

Pair roles: Navigator owns the storyboard; Driver turns each box into a block. Swap at the wiggle break. *(Trio: third checks each box got coded.)*

Differentiation: *Support* — a storyboard with 2 boxes pre-drawn. *Extension / fast-finisher* — a 6-box plan. *Reteach* — code one box together, then let them do the next alone.

Exit check: Pair's picture plan lines up box-to-block with their working sequence. **At-home (optional):** Draw a 4-box picture plan for feeding a pet or watering a plant. ~5 min.
